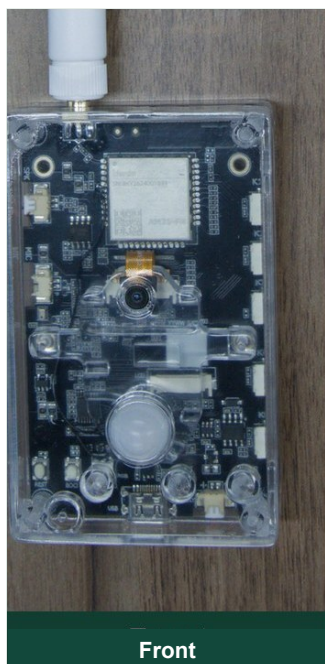


AM36 Electronic Doorbell

Firmware v1.0.0 · 2026-09

Feature Guide

DOOR STATION



Front



Back

15 s

ring and answer window

2 + 5

missed clips + PIR stills in PSRAM

DISPLAY GATEWAY



Front



Back

2.6 Mbit/s

FLRC, 915.12 MHz, 22 dBm max

Opus + AEC

full-duplex voice, echo cancelled

Product Overview

Same firmware, two roles

AM36 Electronic Doorbell is an open-source two-device video doorbell reference built on the Lierda AM36 ESP32-S3 and LR2021 platform. Two identical DK01 kits run the same firmware: one acts as the door station and the other as an indoor display gateway. A visitor press starts a 15-second ring while live door-station video is shown. The gateway can answer into two-way full-duplex voice, or retain an unanswered audio/video record in PSRAM.

The gateway can also start a live view on its own: K5 cycles from viewing and listening, to a two-way call, and back to listening. PIR motion can push still snapshots to the Visitors page. Visitor records and snapshots are volatile and are lost after a reset or power loss.

Repository Documentation

- **README.MD / README_EN.md**: build, controls, and operating instructions.
- **docs/hardware/**: DK01 schematic and Altium design sources.
- **docs/3d-print/**: seven STEP enclosure and mounting models.
- **version.txt / CHANGELOG.md**: firmware version and release history.
- **THIRD_PARTY_NOTICES.md**: source and license attribution.

01 Capabilities and Applications

A two-device doorbell path from a visitor press to live video, voice, and local records.

Product Positioning

A reference design for entrances that need an independent long-range link. It demonstrates one-way live door video, full-duplex voice, volatile visitor records, PIR snapshots, and low-power wake-up without relying on site Wi-Fi.

Core Capabilities

Live door view

The camera produces 288 x 240 frames, rotated onto the gateway's 240 x 320 portrait LCD.

Long-range link

LR2021 FLRC carries video and audio; LoRa CAD is used for low-power wake-up.

Voice and echo control

Two-way full-duplex voice uses Opus, acoustic echo cancellation, and howl suppression.

Local visitor memory

The gateway keeps up to two missed clips and five PIR snapshots in PSRAM.

Typical Applications

Home and detached entrance

See and answer a visitor where cabling or local network access is limited.

Farm, gate, and warehouse

Place the door station at a remote gate and keep the display gateway indoors.

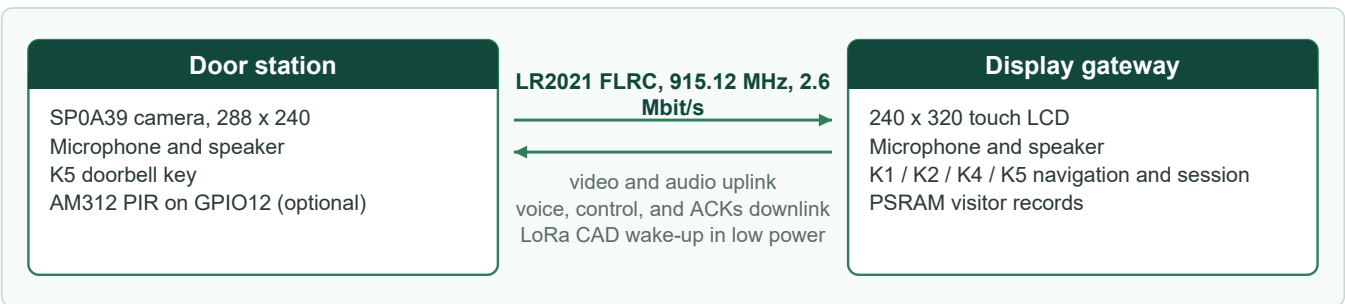
Temporary site

Use the two portable units at temporary offices, cabins, or controlled sites.

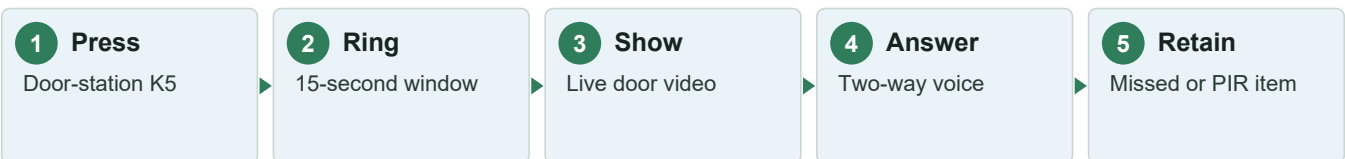
Wireless demonstration

Demonstrate LR2021 video, voice, wake-up, and link statistics on one platform.

System Composition

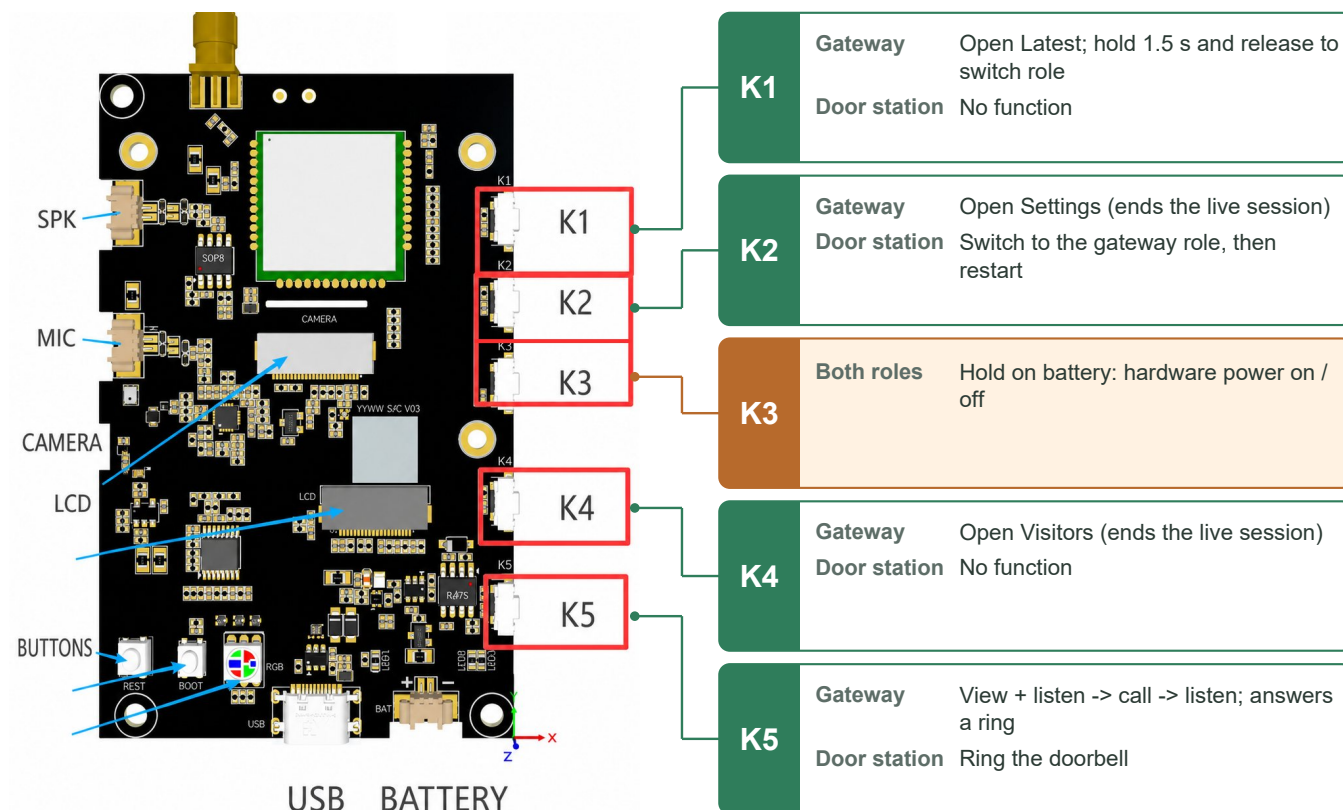


End-to-End Workflow



02 Device Roles, Buttons, and Touch

Two identical DK01 kits run one firmware image; the selected role is stored in NVS.



RST: hardware reset (CHIP_PU) · BOOT: GPIO0 download / boot strap · K1, K2, K4, K5 are read by the application; K3 is a hardware power key.

Role Selection

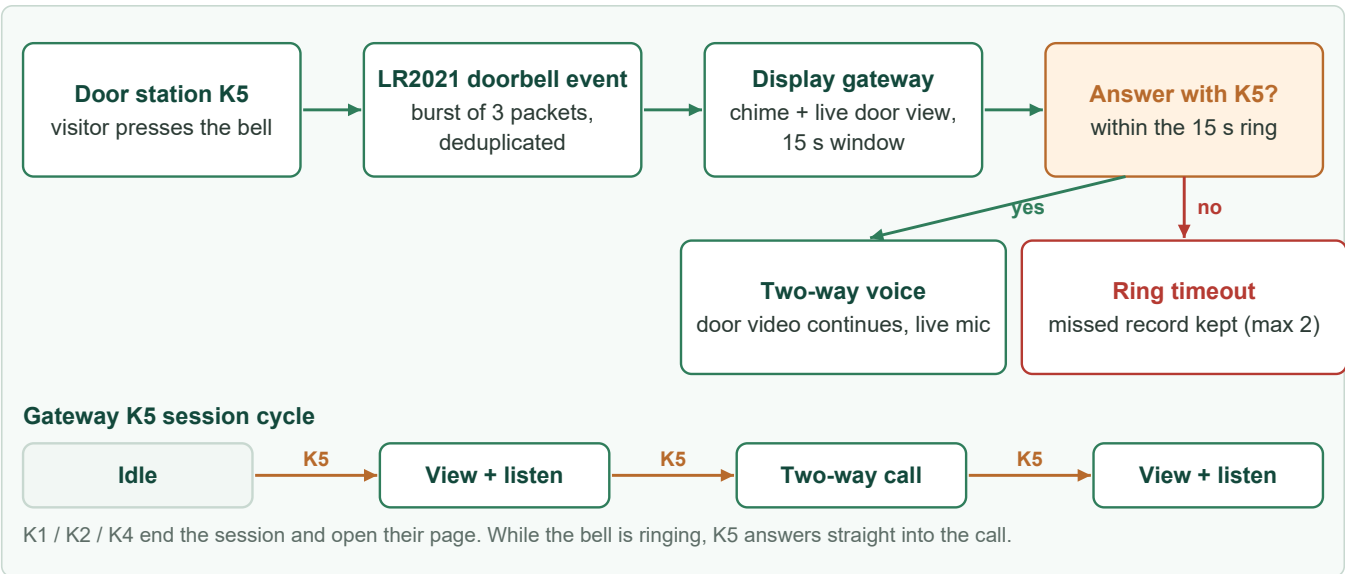
An unconfigured unit starts as the door station. Press door-station K2 to select the gateway role; on the gateway, hold K1 for 1.5 seconds and release to return to the door-station role. The role is stored in NVS and survives restarts.

Touch

Tap on-screen controls directly (the CAPTURE button, switches, visitor tiles). Swipe left or right only among Latest, Visitors, and Settings. During a live session, tapping the picture or pressing a navigation key ends it.

03 Ring, Answer, and the Session Cycle

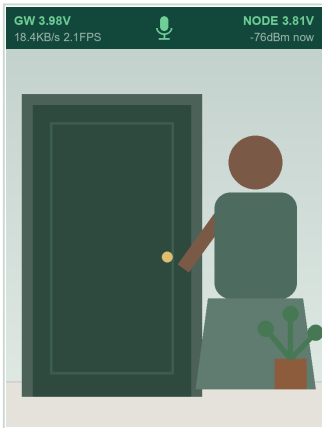
A visitor press opens a 15-second window with the live door view already on screen; one K5 key then runs the whole session.



Ringing: live view + chime



View + listen: muted mic



Two-way call: live mic

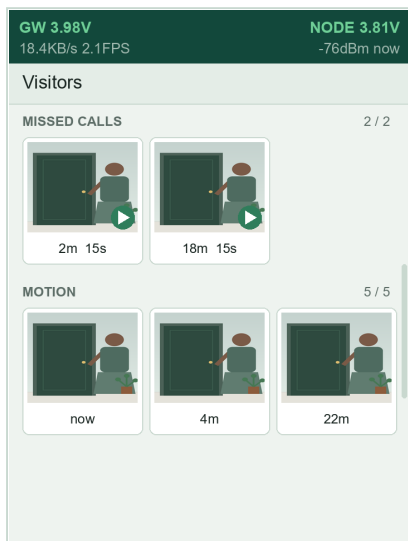
State	Video	Audio	How it ends
Ringing	Door station -> gateway	Chime on the gateway; visitor audio recorded, not played	K5 answers; 15 s timeout keeps a missed record
View + listen	Door station -> gateway	Door mic to gateway speaker; gateway mic muted	K5 starts the call; K1 / K2 / K4 leave
Two-way call	Door station -> gateway	Full-duplex voice with AEC and howl suppression	K5 returns to listening; K1 / K2 / K4 leave

Record rule

Video is one-way from the door station to the gateway; voice is two-way. Only a ring that reaches its unanswered timeout is committed as a missed record; answering or leaving the session discards the pending record. The CAPTURE button on the Settings page runs the same session cycle as K5.

04 Visitor Records and PIR Snapshots

The Visitors page keeps missed-ring clips apart from PIR motion stills; both live only in PSRAM.



Visitors: missed calls, then motion

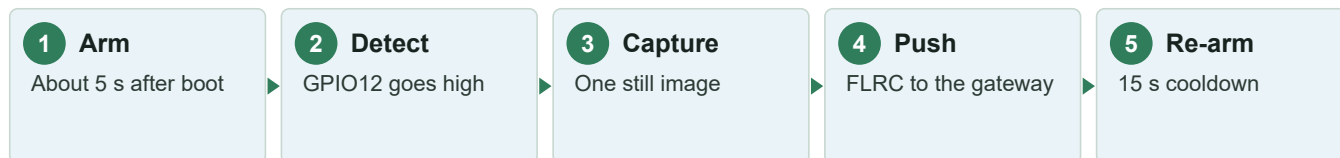
Missed calls

Up to two unanswered doorbell sessions are retained, newest first. Each entry stores the JPEG frames and the Opus audio that arrived during the 15-second ring. A new item evicts the oldest when both slots are full.

Playback and unseen badge

Tap a missed-call tile to replay it with the original frame timing; tap the picture or press any key to stop. A play mark distinguishes clips from stills. Items arriving while Visitors is closed increment the red badge in the status bar; opening Visitors clears it.

PIR Motion Snapshots



Enable

Turn on PIR Motion in Settings. The setting is sent to the door station and stored in NVS on both units. PIR is for the door-station role only, because GPIO12 is also the gateway's touch interrupt.

Arrival

The still is filed under MOTION (at most five; a sixth evicts the oldest) and increments the unseen badge. It does not pull the gateway away from its current page. Tap the tile later to view it full screen.

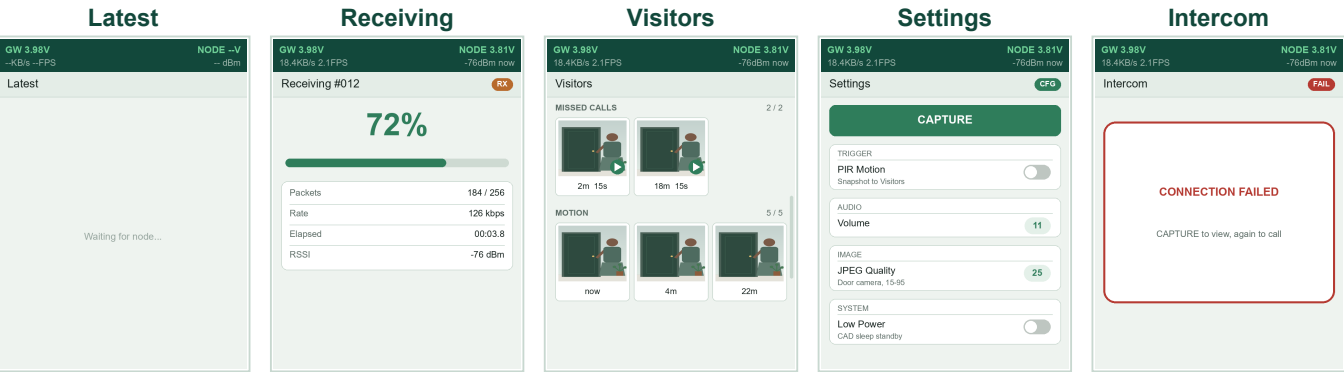
AM312 pin	DK01 connection	Note
VCC	3V3	3.3 V supply.
OUT	GPIO12	Active-high input; door-station role only.
GND	GND	The module and the board must share ground.

Capacity Summary

Item	Capacity	Behavior
Missed call	2	15-second ring window; newest first; audio/video playback.
PIR snapshot	5	Single still image; newest first; tap for full-screen display.
Storage	PSRAM only	All items are lost after a reset or power loss.

05 Gateway Pages and Status Bar

The 240 x 320 interface keeps power, link, event, microphone, and unseen state visible on every page.



Status Bar



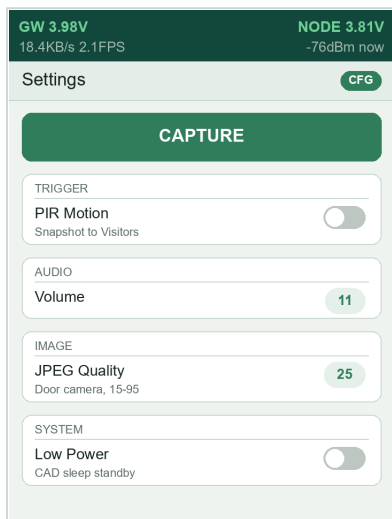
Area	Meaning
1 Row 1 left	Gateway supply voltage, refreshed every minute (green / amber / red by level).
2 Row 1 right	The latest door-station voltage received with a frame or heartbeat.
3 Row 2 left	Link throughput and displayed frame rate over the last frames.
4 Row 2 right	RSSI and the age of the latest event; NODE OFFLINE after 12 minutes without any door-station traffic.
5 Centre	Muted microphone while listening; green live microphone during a call.
6 Red badge	Unseen missed calls and motion snapshots, shown up to 9.

Pages

Page	Purpose
Latest	Waiting state, or the live / full-screen picture. The title bar gives way to the video during a stream.
Receiving	Percentage, packets, rate, elapsed time, and RSSI while the first frame of a session arrives.
Visitors	Missed-call clips and PIR snapshots; tap a tile to play or view.
Settings	CAPTURE button and the PIR Motion, Volume, JPEG Quality, and Low Power controls.
Intercom / Playback	Transient call status (connecting, failed, ended) or a full-screen record playback.

06 Settings

Five controls: live session, motion capture, playback volume, picture quality, and low power.



Settings page

Control	Behavior
CAPTURE	Runs the same session cycle as K5: view and listen, two-way call, back to listening.
PIR Motion	Enables one-shot motion snapshots on the door station.
Volume	Gateway playback level 0 to 15 (default 11), stored in gateway NVS. About 8.5 dB per step: 0 mutes, 11 is near 0 dB, 15 is +32 dB. High levels feed back inside an enclosure.
JPEG Quality	Door-station encoder quality 15 to 95 in steps of 10 (default 25). Higher values: sharper picture, larger frames.
Low Power	Enables the door station's LoRa CAD sleep standby.

Persistence and Delivery

Item	Gateway NVS	Door station
Volume	Yes	Not sent
PIR Motion	Yes, after the ACK	Applied and stored in NVS
Low Power	Yes, after the ACK	Applied and stored in NVS
JPEG Quality	Yes, after the ACK	Applied and stored in NVS

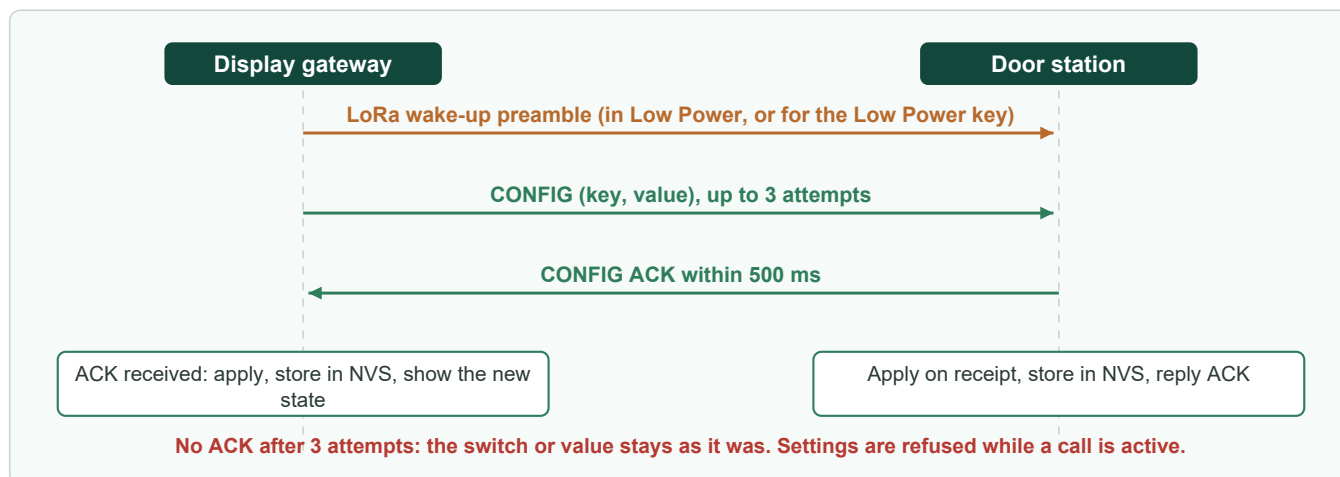
Acknowledged changes

PIR Motion, Low Power, and JPEG Quality change only after the door station's ACK; if delivery fails, the switch or value stays as it was (sequence below).

Fixed RF configuration and regional compliance

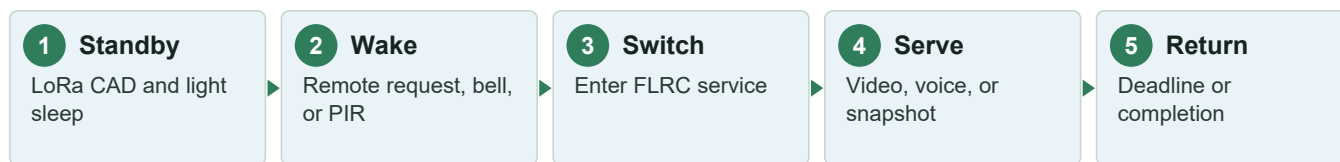
915.12 MHz, 2.6 Mbit/s FLRC, up to 22 dBm, no runtime frequency selector. Confirm that this frequency and power are permitted in the deployment region.

Delivery Sequence



07 Low-Power Operation

The door station waits in LoRa CAD, wakes on demand, and uses FLRC only for service traffic.



Wake sources

- Gateway live-view request (long LoRa preamble, then FLRC)
- Door-station K5 bell press
- GPIO12 PIR high level

A bell press or PIR event wakes the door station locally; it then pushes the event or the snapshot on its own.

Hard deadline

A CAD wake request opens a fixed 15-second FLRC window; ordinary traffic does not extend it. Starting a two-way call opens a fresh 15-second deadline. At expiry, unfinished service stops and the door station returns to CAD.

Phase	Door station	Display gateway
CAD standby	LR2021 scans for the LoRa wake pattern; the ESP32-S3 sleeps between scans and the camera is powered down.	Stays in continuous FLRC receive so an unrequested PIR push can arrive at any time.
Remote request	Wakes on the preamble, switches to FLRC, and serves frames inside the window.	Sends the long-preamble wake-up, then the frame requests.
Local bell / PIR	Wakes locally and pushes the doorbell event or the snapshot.	Rings, or files the snapshot under MOTION.
Deadline	Stops the transfer or the voice session and returns to CAD standby.	Ends the session on its side at the next key press.

Configuration

Low Power is changed on the gateway Settings page, acknowledged by the door station, and stored by both units. With Low Power off, the door station stays in FLRC receive and the 15-second deadline does not apply. The USB serial console does not survive light sleep, so serial output pauses during CAD standby.

08 Getting Started and Verification

Build once, flash both identical units, select their roles, and verify the complete visitor path.

Step	Action
1	Prepare two L-LRMAM36-FANN4-DK01 kits and USB cables.
2	Use ESP-IDF v5.5.3; run <code>idf.py set-target esp32s3</code> , then <code>idf.py build</code> .
3	Run <code>idf.py -p COMx flash</code> for each unit with the same build.
4	Leave one unit in the default door-station role; press K2 on the other unit to make it the gateway.
5	Press door-station K5 and verify the gateway chime and live door image.
6	Answer with gateway K5; press it again to return from the call to listening.
7	Let a second ring time out and verify it under Visitors.
8	Optionally wire an AM312, enable PIR Motion, and verify a motion snapshot.

Verification Checklist

Check	Expected result
Role persistence	The selected role remains after a restart.
Live path	Door video, listening, and two-way voice all operate.
Visitor memory	At most two missed clips and five PIR stills, newest first.
Power cycle	Visitor media is cleared because it is stored only in PSRAM.
Low power	With Low Power on, a gateway request still wakes the door station and the window closes after 15 s.

Troubleshooting

Symptom	What to check
The gateway does not ring	One unit must be the door station and the other the gateway (two units in the same role do not talk). Shorten the distance or check the antennas. A second press during a ring session is ignored on purpose.
NODE OFFLINE in the status bar	No door-station traffic for 12 minutes. A door station in Low Power sends no periodic heartbeat, so this appears after 12 idle minutes and clears on the next event.
Echo or howling during a call	Lower Volume a few steps (each step is about 8.5 dB; 11 is close to 0 dB gain), increase the distance between the units, and keep the speaker from facing the microphone.
PIR never triggers	Check the AM312 wiring (OUT to GPIO12, common ground), that PIR Motion is on and was acknowledged, and allow about 5 s after boot and 15 s between triggers.
Door picture looks blocky	Raise JPEG Quality in Settings while the door station is reachable; the value takes effect from the next frame and is kept by the door station.
Visitor records disappeared	Expected: records and snapshots live only in PSRAM and are cleared by any reset or power loss.